

Lecture-27

Mathematics 1 (15B11MA111)

CO [C105.5]

Module: Differential Equations
Topic: Particular Integral

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Content

- ☐ Special cases for particular integral
- ☐ General formula for particular integral
- ☐ Practice questions

Rules for Finding the Particular Integral

□ Case 3: When $r(X) = X^\alpha$, $\alpha > 0$

The particular Integral of $F(D)y = X^\alpha$, $\alpha > 0$, is given as

$$y_p(x) = [F(D)]^{-1} X^\alpha ,$$

Expand $[F(D)]^{-1}$ as an infinite series in ascending powers of D and then operate on X^α .

Example: Find the particular Integral of the following differential equation:

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4.$$

Solution: Given differential equation is $(D^2 + D)y = x^2 + 2x + 4$. The particular integral is given as

$$\begin{aligned} y_p(x) &= \frac{1}{F(D)} (x^2 + 2x + 4) = \frac{1}{(D^2 + D)} (x^2 + 2x + 4) \\ &= \frac{1}{D(D+1)} (x^2 + 2x + 4) \\ &= \frac{1}{(D)} (1 + D)^{-1} (x^2 + 2x + 4) \\ &= \frac{1}{D} (1 - D + D^2 - \dots) (x^2 + 2x + 4) \\ &= \frac{1}{D} [(x^2 + 2x + 4) - (2x + 2) + 2] \\ &= \frac{x^3}{3} + 4x \end{aligned}$$

Rules for Finding the Particular Integral (Contd..)

□ **Case 4: When $r(X) = e^{\alpha X}h(X)$, $\alpha > 0$**

The particular Integral of $F(D)y = e^{\alpha X}h(X)$, $\alpha > 0$, is given as

$$\begin{aligned}y_p(x) &= [F(D)]^{-1}e^{\alpha X}h(X) \\&= e^{\alpha X}[F(D + \alpha)]^{-1}h(X) \\&= e^{\alpha X} \frac{1}{F(D+\alpha)} h(X)\end{aligned}$$

Example: Find the general solution of the following differential equation: $16 \frac{d^2y}{dx^2} + 8 \frac{dy}{dx} + y = 48xe^{\left(-\frac{x}{4}\right)}$

Solution: Given differential equation is

$$(16D^2 + 8D + 1)y = 48xe^{\left(-\frac{x}{4}\right)}.$$

The auxiliary equation is given as

$$16m^2 + 8m + 1 = 0.$$

$$\text{On solving, } m = -\frac{1}{4}, -\frac{1}{4}.$$

Hence the complimentary function is given by **CF** = **(A + Bx) e^($-\frac{x}{4}$)**.

Now the particular integral is given as

$$\begin{aligned}y_p(x) &= \frac{1}{F(D)} 48xe^{\left(-\frac{x}{4}\right)} = 48 \frac{1}{(16D^2 + 8D + 1)} (xe^{\left(-\frac{x}{4}\right)}) \\&= 48e^{\left(-\frac{x}{4}\right)} \frac{1}{\left(16\left(D - \frac{1}{4}\right)^2 + 8\left(D - \frac{1}{4}\right) + 1\right)} (x) \\&= 48e^{\left(-\frac{x}{4}\right)} \frac{1}{(16D^2)} (x) = 3e^{\left(-\frac{x}{4}\right)} \frac{x^3}{6}\end{aligned}$$

The general solution is therefore,

$$y(x) = CF + PI$$

$$= (A + Bx) e^{\left(-\frac{x}{4}\right)} + 3e^{\left(-\frac{x}{4}\right)} \frac{x^3}{6}$$

Rules for Finding the Particular Integral (Contd..)

□ **Case 5: When $r(X) = X.h(X)$,**

The particular Integral of $F(D)y = X.h(X)$, is given as

$$y_p(x) = [F(D)]^{-1} X.h(X)$$

$$= X \frac{1}{F(D)} h(X) + \frac{d}{dD} \left(\frac{1}{F(D)} \right) h(X)$$

$$= X \frac{1}{F(D)} h(X) - \left(\frac{F'(D)}{[F(D)]^2} \right) h(X)$$

Example: Solve the differential equation $y'' - y = x \sin x$

Solution: Given differential equation is

$$(D^2 - 1)y = x \sin x.$$

The auxiliary equation is given as

$$m^2 - 1 = 0.$$

On solving, $m = -1, 1$.

Hence the complimentary function is given by

$$CF = Ae^x + Be^{-x}.$$

Now the particular integral is given as

$$\begin{aligned}y_p(x) &= \frac{1}{F(D)} x \sin x = \frac{1}{D^2 - 1} (x \sin x) \\&= x \frac{1}{(D^2 - 1)} \sin x - \left(\frac{2D}{[D^2 - 1]^2} \right) \sin x \\&= x \frac{1}{-2} \sin x - \left(\frac{2D}{4} \right) \sin x \\&= -x \frac{\sin x}{2} - \left(\frac{2 \cos x}{4} \right)\end{aligned}$$

The general solution is therefore, $y(x) = CF + PI$

$$y(x) = Ae^x + Be^{-x} + \left(-x \frac{\sin x}{2} - \left(\frac{2 \cos x}{4} \right) \right)$$

General Rule for Particular Integral

The particular Integral of the differential equation

$F(D)y = (D - \alpha)y = r(x)$ is given as

$$\begin{aligned} y_p(x) &= \frac{1}{(D-\alpha)} r(x) \\ &= e^{\alpha x} \int e^{-\alpha x} r(x) dx \end{aligned}$$

Example: Find the particular integral of the differential equation $y'' + y = \sec x$

Solution: Given differential equation is $(D^2 + 1)y = \sec x$.

The particular integral is given as

$$\begin{aligned} y_p(x) &= \frac{1}{(D^2 + 1)} \sec x = \frac{1}{(D - i)(D + i)} (\sec x) \\ &= \frac{1}{2i} \left[\frac{1}{(D - i)} - \frac{1}{(D + i)} \right] \sec x \end{aligned}$$

$$\begin{aligned} \text{Now, } \frac{1}{(D - i)} \sec x &= e^{ix} \int e^{-ix} \sec x \, dx \\ &= e^{ix} \int (\cos x - i \sin x) \sec x \, dx \\ &= e^{ix} \int (1 - i \tan x) \, dx \\ &= e^{ix} (x + i \log (\cos x)) \end{aligned}$$

$$\text{Similarly, } \frac{1}{(D + i)} \sec x = e^{-ix} \int e^{ix} \sec x \, dx = e^{-ix} (x - i \log (\cos x))$$

Therefore,

$$\begin{aligned}y_p(x) &= \frac{1}{(D^2+1)} \sec x = \frac{1}{2i} \left[\frac{1}{(D-i)} - \frac{1}{(D+i)} \right] \sec x \\&= \frac{1}{2i} e^{ix} (x + i \log (\cos x)) - \frac{1}{2i} e^{-ix} (x - i \log (\cos x)) \\&= x \sin x + \cos x \log (\cos x)\end{aligned}$$

Practice Questions

1. Solve the differential equation $y'' - 2y' + y = x e^x \sin x$.

(Ans. $y(x) = (A + Bx)e^x - e^x(x \sin x + 2 \cos x)$)

2. Find the particular integral of the following *differential equation*

$$y'' - 3y' + 2y = x e^{3x} + \sin 2x.$$

(Ans. P.I. = $e^{3x} \left(\frac{x}{2} - \frac{3}{4} \right) + \frac{1}{20} (3 \cos 2x - \sin 2x))$)

3. Solve $(D^2 - 1)y = x \sin 3x$

(Ans. $y(x) = (Ae^x + Be^{-x}) - \frac{1}{10} (x \sin 3x + \frac{3}{5} \cos 3x))$)

THANK YOU